



AlLoRa: Empowering environmental intelligence through an advanced LoRa-based IoT solution

Benjamín Arratia*, Erika Rosas, Carlos T. Calafate, Juan-Carlos Cano, José M. Cecilia, Pietro Manzoni

DISCA-GRC, Universitat Politècnica de València, Valencia, Spain

ARTICLE INFO

Keywords:

LoRa
IoT
Mesh
Environmental intelligence
Green connectivity

ABSTRACT

Environmental intelligence aims to improve the decision-making process for high social and environmental value ecosystems. To this end, data are collected using different sensors to allow monitoring of different variables of interest. Typically, these ecosystems cover a large geographical area, with spots of low or no connectivity, preventing their monitoring in real time. In this work, we propose *AlLoRa* (Advanced Layer LoRa), a modular, low-power, long-range communication protocol based on LoRa, that allows monitoring of remote natural areas. *AlLoRa* has been evaluated and tested in an operational oceanographic buoy that has been deployed to address the specific environmental crisis of the Mar Menor lagoon in southeastern Spain - a region spanning 135 Km² currently undergoing severe eutrophication process.

Our results reveal that *AlLoRa* offers good performance regarding transfer time, power consumption, and range. The throughput ranged from around 2 kbps with SF7 to approximately 300 bps with SF11; the power consumption per kilobyte transmitted varied from 395 µWh to 428 µWh depending on the specific device used. The Mesh mode test successfully maintained communication between nodes over 20.33 km. Further tests in various configurations under challenging conditions validated the mesh forwarding approach. Despite tripling the distance, the system maintained reliable data transfer, improving speeds from the original point-to-point setup.

1. Introduction

Environmental sensing provides mechanisms for efficiently treating water, waste, or emergencies with which to face the new challenges of globalization and climate change [1]. This sector evolves into the so-called Environmental Intelligence (EI), where information about a particular region or process is collected for decision-makers' benefit using more than one inter-related source. The proliferation of sensors increasingly changes how we relate to and manage the environment, providing data to answer scientific questions and generating social, cultural, and political impacts.

The main data generation engine for EI is the Internet of Things (IoT), where numerous devices interact with our environment, monitoring it and even autonomously taking actions [2]. The potential scalability of IoT can facilitate the creation of a Global Environmental Sensor Web [3], offering an analytical tool for a better understanding of our planet and enabling informed ecological decisions [4].

However, implementing and adopting IoT systems in natural environments encounter numerous challenges, from communication and network architectures to data analysis [5]. These environments often

span vast geographical areas with limited connectivity [6], presenting a significant challenge in designing novel approaches for integrating, operating, and managing large-scale IoT networks.

To address these challenges, this research introduces **Advanced Layer LoRa (AlLoRa)**, a modular communication protocol tailored for EI applications. *AlLoRa* enhances IoT devices' connectivity and data transmission capabilities in remote and challenging natural environments, promoting the scalability and integration of environmental sensing networks. The protocol's design prioritizes energy efficiency and long-range capabilities to ensure sustainable operation and extensive coverage. Its adaptability to various hardware configurations and support for mesh networking further facilitates the deployment of expansive sensor networks in ecologically sensitive areas.

Most existing IoT solutions are built on proprietary systems, creating dependencies on external entities and often come with the added complexity and cost of tailoring parts of the communication protocol.

This article is an extension of [7], where a preliminary LoRa-based connectivity solution for environmental intelligence was presented. In this work, we elaborate on *AlLoRa*; a long-range, low-power, and

* Corresponding author.

E-mail address: baarruri@disca.upv.es (B. Arratia).

modular communication protocol based on LoRa, which facilitates monitoring remote natural areas. Several new features have been developed, improving the overall capabilities of the toolbox to make it applicable in more contexts and compatible with more hardware setups.

ALoRa is developed within the context of the SMARTLAGOON project, aiming to develop a Digital Twin of Europe's largest salt-water coastal lagoon, the Mar Menor in Spain [8]. This lagoon is a hyper-saline coastal ecosystem with high environmental value. Recent changes in water quality have caused social alarm [9], sparking several initiatives. Despite the vulnerability and complexity of these ecosystems, there has been limited development of novel techniques that can provide real-time monitoring, analysis, and management of these critical resources.

This research aims to advance IoT systems' capabilities for environmental monitoring by offering a robust, scalable, and efficient alternative to current solutions. We aim to demonstrate the application of *ALoRa* for real-time data acquisition, enhance environmental understanding and conservation efforts, and validate the protocol's effectiveness in achieving long-distance communication with low power consumption.

ALoRa enables data collection on water temperature, chlorophyll-a, and oxygen concentration from sensors located on an oceanographic buoy. This capability for real-time data collection and analysis significantly enhances our understanding of the environment and enables proactive decision-making for environmental conservation.

In this oceanographic and environmental context, energy consumption is critical for several reasons, including the need to minimize the environmental impact of the solutions developed to conserve the environment and ensure the systems' autonomy. The scalability of these oceanographic systems to host a larger number of nodes is also relevant, as it allows for extending geographical coverage. Moreover, long-distance connection ranges are necessary in oceanographic contexts to retrieve and transmit collected data to a receiving node.

ALoRa addresses these issues and allows generic content transfer using LoRa technology. It is designed to be compatible with a wide range of devices, facilitating the adoption and deployment of edge devices that require transferring large data files over long distances with low power consumption. It also supports mesh mode, which enables message forwarding between nodes to extend communication range. Our experimental results reveal that long distances can be achieved within a reasonable budget, showing energy consumption figures very attractive for this kind of environmental intelligence system.

The rest of the paper is structured as follows: Section 2 shows the critical technologies behind *ALoRa* and related work. Section 3 details the design and logic of the communication protocol. The results of the tests performed using our prototype are displayed and analyzed in Section 4. Finally, Section 5 presents our conclusions and directions for future work.

2. Background and related work

We propose a communication architecture and protocol based on the LoRa technology [10]. LoRa is a Low Power Wide Area Network (LPWAN) technology that uses a proprietary spread spectrum modulation technique derived from the chirp spread spectrum (CSS) to achieve long-range wireless transmissions at extremely low power. Chirps make LoRa resilient to noise, fading, and interference. LoRa has been widely proposed for IoT solutions since it allows long-range communications, from 2 to 5 km in urban areas and 15 km in suburban areas [11]. It allows data rates from 250 B/s to 5.5 B/s using unlicensed sub-GHz ISM band, 868MHz frequency band in Europe, 915MHz in America, and 433 in Asia. There are four parameters that LoRa defines: Bandwidth (BW), Spreading Factor (SF), Carrier Frequency (CF), and Code Rate (CR). Their selection defines a trade-off between transmission range and data rate. The bandwidth determines the data transmission rate. Its

frequency range of 125 to 500 kHz allows for broad design options for several applications. The SF defines the signal's processing gain and, therefore, airtime and latency of packets. It is defined as the base-2 logarithm of the number of chirps per symbol, and it runs from SF7 to SF12, with higher SF values providing longer transmission ranges but slower data rates and longer airtime. The CR indicates the signal's capability to correct errors. With values ranging from CR4/5 to CR4/8, the communication system can withstand more errors and interference with a higher CR, but at the expense of a lower data rate.

Recent research has identified LoRa as a key technology to address the connectivity requirements of IoT systems providing cost-efficient long-range coverage [12]. Tangible implementations of LoRa in *smart* applications have demonstrated successful deployment. However, these applications come with issues and challenges, ranging from concerns about security, scalability, and network lifetime [13,14].

2.1. LoRa for environmental intelligence

The straightforward approach of using the benefits of the LoRa technology is the implementation of LoRaWAN, which is at the MAC layer of the OSI Reference model and is an open specification standard managed by LoRa Alliance [15]. It uses a star-of-stars topology, where a gateway connects to IoT devices using a single-hop link. LoRaWAN has been extensively proposed for sensing and monitoring environmental parameters with low-cost and long-range communications [16–18]–[19]. However, the LoRaWAN star-of-stars topology is not suitable for challenged networks, as a single hop is not enough to achieve long-range coverage in environments with poor infrastructure and a heterogeneous mix of resource-constrained nodes, as is the case of the Smartlagoon European project. Centelles et al. [20] elaborate on three representative scenarios that would benefit from a more flexible topology than LoRaWAN. One of them applies to our case scenario, which is extending the coverage beyond the infrastructure reach. Using LoRaWAN implies the installation of more gateways in a geographic area where we want to intervene as little as possible.

In the literature, we have found approaches that improve the geographical coverage of LoRaWAN. We focus our analysis on software improvements that do not require modification of commercially available chips. Cotrim and Kleinschmidt [21] survey recent studies that proposed multihop LoRaWAN networks, thus forming wireless mesh networks. Some solutions use the LoRaWAN characteristics, while others use only the LoRa physical layer. The former solutions generally introduce multi-hop networking using existing protocols such as Ad hoc On-Demand Distance Vector (AODV) [22] or Destination-Sequenced Distance Vector (DSDV) [23]. Similarly, Prade et al. [24] propose Multi-LoRa for large-scale IoT deployments, using smart agriculture in farms with hundreds of square kilometers. The authors propose a routing algorithm using a routing table and a neighbor table. To maintain tables, messages need to be exchanged between neighbors periodically, as proposed by Solé et al. [25] in the LoRa Mesh Library. Our solution avoids using routing protocols that maintain tables and exchange routing protocol messages to favor a low-energy solution, excluding unnecessary overhead and complexity.

Linear multihop communication increases network coverage with a more straightforward approach and has been proposed in underground scenarios for urban water infrastructure management, like tunnels and pipelines. For example, Abrardo and Pozzebon have used a LoRa linear multi-hop network for aqueducts in Siena, Italy [26], and Ebi et al. [27] deployed the wireless sensor network in Fehraltorf, in Zurich, Switzerland. In underground facilities, high signal path attenuation is one crucial challenge, together with packet loss, and solutions require using a larger number of intermediary devices.

To deploy a low-energy solution, our approach uses only the physical layer of LoRa. Among similar approaches, Mai and Kim [28] proposed a tree-based protocol built by exchanging packets between the LoRa nodes and the sink node. A time slot channel is assigned

to each tree link, producing a collision-free approach. Another tree-based solution is MLoRa, proposed by Feng et al. [29], which uses a spanning tree variation to enable multi-hop over LoRa and extend end devices' lifetime. Tran et al. propose a two-hop real-time LoRa protocol [30] focused on dynamic configurations. The protocol is built with a tree topology and requires slot scheduling, topology maintenance, and updating the slot schedule. Acknowledging that these solutions produce high overhead, Ahmar et al. [31] propose Smart-hop. This approach proposes a low-overhead routing, eliminating the complexity and overhead of traditional multi-hop mesh solutions. In Smart-hop the focus is on low latency communications, achieved by selectively replacing high SF links with chains of lower SF links. Similarly, MRT-LoRa [32] focuses on maintaining a shorter time in the air at each hop. However, these solutions still maintain a tree-based topology for forwarding messages.

LoRaHop proposes to perform several flooding rounds during which end-devices communicate on multiple channels without the need to establish and maintain routers or a network hierarchy which simplifies network formation at a cost of energy consumption [33]. Hun and Kim [34] proposed a LoRa protocol that supports mesh and time-division multiple access (TDMA). The mesh networks dynamically connect end nodes and self-configure routing paths using a flooding algorithm, then a mesh table is prepared for each node based on the RSSI of beacon messages. Wu and Lieberherr have studied extensively collision mitigation methods that include multichannel communication and adaptive random backoff delays, and propose to organize a mesh LoRa network as a rooted spanning tree that synchronizes duty cycles disseminating information of the next round in the preceding duty cycle [35]. Similarly to work in [36], we use a polling-based approach, where devices listen to gateway requests. This approach avoids collision and is efficient with several sensor nodes. Unlike these solutions, our intermediary nodes do not spontaneously build routes and forward messages. Such an approach is appropriate in scenarios with poor infrastructure, as the one we found in Mar Menor, where the sensors are sparse in the geographical region.

Another recent approach for mesh and multi-hop LoRa-based networks is the one proposed by Berto et al. [37]. Their solution is completely gateway-free, enabling peer-to-peer communication. The authors have a use case that differs from the one we target without the requirement of carrying all the data to one particular node for monitoring or processing.

In the context of rivers and bays, Guibene et al. [38] deployed a buoy on the Liffey River in Dublin city center to monitor river depth, air, water temperature, barometric pressure, and solar/battery performance. The authors used the physical layer of LoRa using SF12 and a bandwidth of 125 KHz for eight months and obtained an overall error of 1.78%. More recently, the work proposed by Cecilio [39], called AQUAMESH, aims at collecting real-time information in environments where LoRaWAN point-to-point communication cannot provide sufficient coverage. The authors adapt the LoRaWAN stack protocol to support mesh capabilities and use TDMA to extend the node lifetime. Another real application of LoRa communication in a hydrological setting was presented by Nordin et al. [40], where the authors deployed four sensors and a gateway in the Tasik Chini lake in Pahang, Malaysia. LoRaWAN has also been the key technology to the proposal of Sendra et al. [41] that build a sensor network comprised of a buoy that monitors water turbidity, salinity levels, water temperature, presence of oils and fuels, and meteorological parameters.

Long-range inter-vessel low-cost communication has been studied in [42] for building coastal IoT services. Measurements, such as water temperature, wave period, and height or direction of current, are to be exchanged in a mesh network. The solution proposed utilizes the pyMesh library developed by Pycom [43], which is, in turn, based on OpenThread developed by Google. The latter manages the routing algorithms. The authors propose an architecture with two levels, the first level an opportunistic LoRa-mesh network and the second

level a social network. The mesh network requires a watchdog that constantly monitors the network status.

Monitoring large aquatic environments is a research problem that has been stated recently by Msaad et al. in [44], oriented to give support to the so-called Ocean of Things deploying a distributed network of low-cost sensors for real-time data acquisition to support the sustainable management of aquatic environments. This preliminary analysis presents two possible scenarios: one with static sensor platforms, such as the buoy we deployed in the Mar Menor, and another that includes moving platforms, such as vessels or boats. No results are presented in the mentioned work. *AlLoRa* shares the context of these solutions, emphasizing achieving long-range coverage at a low energy cost, building a modular protocol implementation to facilitate the deployment in heterogeneous devices while proposing a simple and efficient solution.

3. Proposed solution

In this section, the *AlLoRa* content transfer protocol is described in terms of its modular architecture, its communication protocol, and the format of exchanged packets. Moreover, we explain the mesh mode and additional services of *AlLoRa*. The open-source software, licensed under GPL-3.0, can be accessed at the GitHub repository indicated in [45].

3.1. *AlLoRa* Architecture

AlLoRa is a half-duplex protocol based on a classical stop-and-wait ARQ scheme, where one or various *Source Nodes* wait for commands from a *Requester/Gateway Node* and answer with the requested data. It offers two operation modes, i.e. *point-to-point* and *mesh* modes. The latter extends the former by increasing the coverage area, and using intermediary nodes to forward packets until reaching the destination node. The mesh mode is activated if the *Source node* does not respond after a certain number of queries that is specified by the user.

AlLoRa is developed using a series of interchangeable modules to simplify its deployment with a wide range of devices, giving access to the content transfer protocol and its combined capabilities. *AlLoRa* packets contain the content transmitted by the *Source Node*, which is split into chunks (blocks of bytes) that are sent using the protocol. The content transfer protocol is symmetrically structured, as shown in Fig. 1.

In this architecture, a *Source Node* receives a *File* object, typically in JSON format, from a *Datasource* (e.g., an oceanographic buoy). It uses a *Connector* to access a LoRa data channel to send the packets. On the other side, a *Requester Node* is built around a *Digital Endpoint*, an object that receives and processes or stores the information coming from a *Source Node*. It also uses a *Connector* to access LoRa.

The *AlLoRa* architecture is composed of several key building blocks. The *Node* manages the communication logic for the *AlLoRa* protocol. It can function as a *Source Node*, a *Requester Node*, or a *Gateway Node*. The *Source Node* waits and listens for requests to send chunks of content until it finishes and is ready to send another file. The *Requester Node* requests information from a *Source Node* and listens for responses. A *Gateway Node* is similar to a *Requester Node*, but can manage multiple *Digital Endpoints*. Fig. 2 depicts a *Gateway Node*, where it communicates with three *Source Nodes*.

The *Connector* bridges LoRa, providing access to a LoRa channel for a *Node*. A *Connector* aims to make *AlLoRa* available to as many types of devices as possible, considering that some devices may have embedded LoRa capabilities, while others may not.

The *AlLoRa File* is a critical component that manages the content to be sent or received by *Nodes*. It can be initialized with data (a byte array) to be used by the *Source Node* to transmit its chunks. Alternatively, it can be instantiated as a container on the *Requester's* side to gather the chunks and assemble them to obtain the original content.

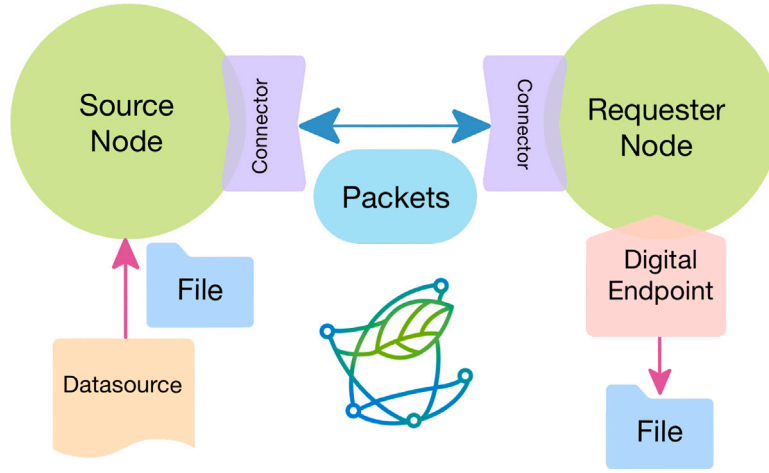


Fig. 1. The various modules constituting the *ALoRa* architecture.

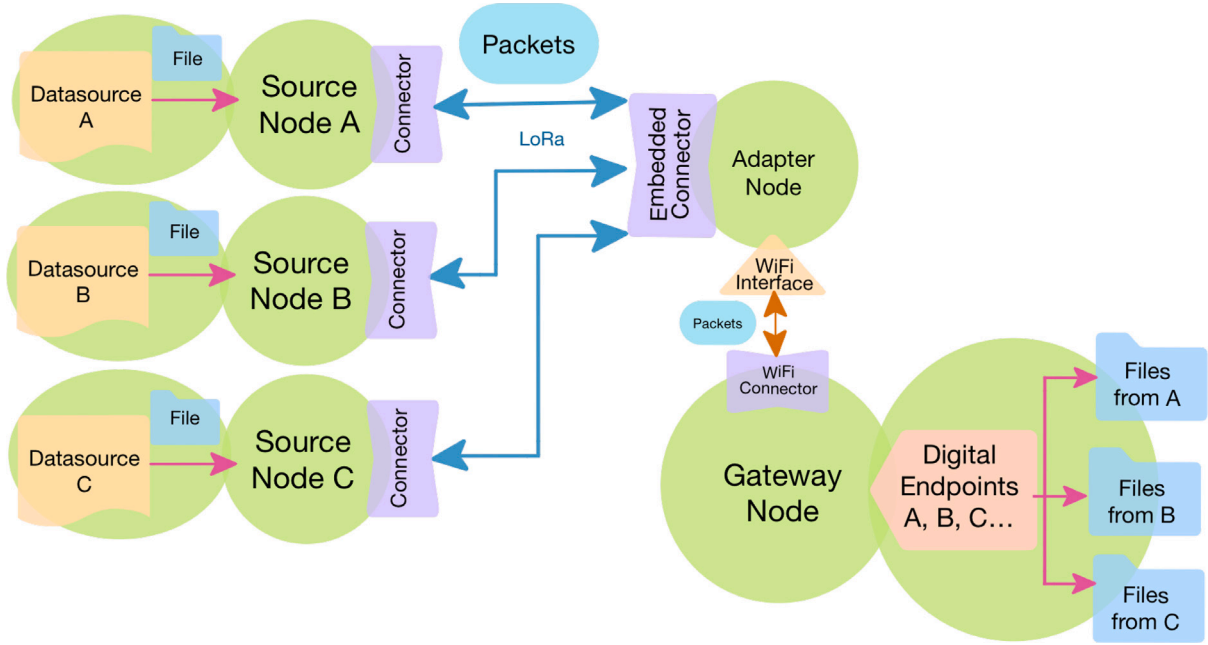


Fig. 2. *ALoRa* architecture with a Gateway communicating with 3 Source Nodes using a Wi-Fi adapter Node.

The *Packet* acts as the envelope for sending the content through LoRa. It structures the message to be sent and can load the data received to check that the content was correctly received (with a checksum). The packet structure is described in Section 3.2.

The *Digital Endpoint* represents a *Source* companion on the *Requester*'s side. It is used by the *Requester* or *Gateway Node* to communicate with the designated *Source*. The object contains the MAC Address of the endpoint to communicate with and manage the transfer state. It also handles the *File* to be received.

The *Datasource* is designed to provide *ALoRa* files to a *Source Node* by facilitating interactions with the data source for the node (usually sensors). It contains methods to read and store data, as well as to retrieve it when needed.

Finally, *Adapters* and *Interfaces* represent advanced components within the *ALoRa* architecture. An *Adapter*, functioning as a distinct type of *Node*, facilitates communication between an *ALoRa* network and a device that does not possess native LoRa capabilities. It manages an *Interface* and a *Connector*. The *Interface* standardizes the communication process across different hardware and communication methods,

enabling the *Adapter* to interact seamlessly with various devices and protocols, such as Wi-Fi or Serial connections.

For example, in a typical *ALoRa* communication setup, a *Source Node* might use a Wi-Fi or Serial *Connector* to communicate with a corresponding *Interface* on an *Adapter Node*. The *Adapter* then utilizes a LoRa *Connector* to transmit the packets over LoRa. On the other side, a *Requester* or *Gateway Node*, also equipped with a LoRa *Connector*, receives the data. This flexibility allows for diverse hardware integrations within *ALoRa*, as demonstrated in Fig. 2.

This modular approach ensures that *ALoRa* can be easily adapted and deployed in various scenarios, ranging from simple point-to-point setups to complex networks involving multiple types of devices.

3.2. Protocol packet structure

The *ALoRa* packets are the basic data unit sent and received through the LoRa channel. They were designed to maximize the amount of content (or chunk size) sent but also to ensure the right reception of the package by the *Requester* node. The *ALoRa* packet has a maximum size of 255 bytes, the maximum payload of a LoRa message.

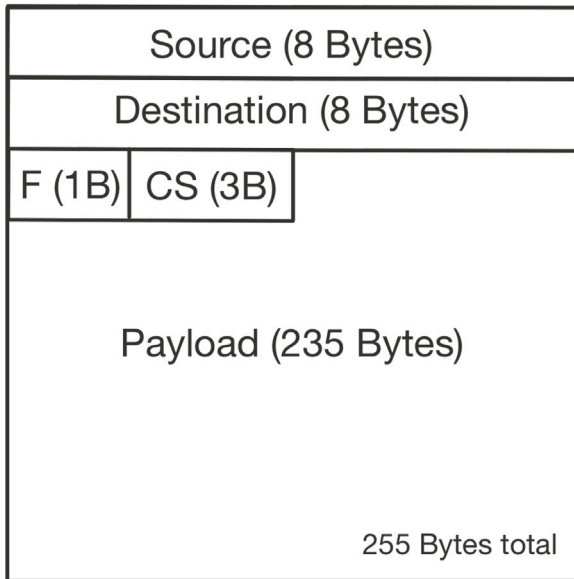


Fig. 3. Point-to-point Packet.

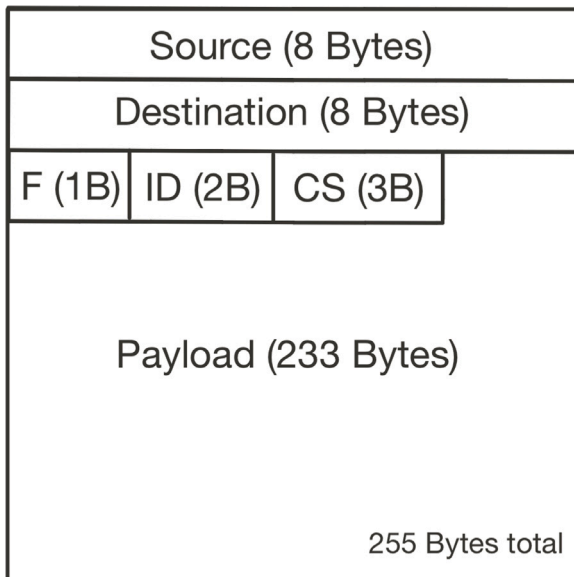


Fig. 4. Mesh-mode Packet.

Figs. 3 and 4 show the format of the *AlLoRa* packets for both point-to-point and mesh communication modes, respectively. The header size varies depending on the communication mode, but both have a standard header of 20 bytes. The first 16 bytes contain the first eight characters of the MAC Address of the source and destination nodes. 1 Byte is used for the message's command and flags (explained below). Three bytes are reserved for a checksum of the content used to check if the content arrived correctly or has some corruption. We use the rightmost three bytes of the SHA-256 hash digest, computed over the data to be sent. Finally, if the system works in *Mesh mode*, an additional 2 bytes store a message ID. The ID is a random number between 0 and 65.535% (the range of values that can be represented in binary using 2 bytes), which is used to manage the retransmissions and avoid chunk duplication. Therefore, the maximum payload of the Point-to-Point mode packet is 235 bytes, while the Mesh mode packet has 233 available for payload.

The Flag byte is structured as follows (see Fig. 5).

- Command bits: 2 bits that combined represent one of the four types of commands detailed in Table 1.
- Mesh bit: Indicates whether or not the message should be forwarded.
- Sleep bit: Determine whether a node should sleep before sending the message. If this bit is activated, the device will wait before forwarding the message. If it is set to false, the node will immediately forward the message.
- Hop bit: If the message was forwarded at some point, then this bit is set to true. Section 3.3.2 provides more details on the Mesh, Sleep, and Hop bit.
- Debug hop bit: Indicates that the message in question is in “debug hop mode”, used only for research purposes. It allows the system to overwrite the content to be sent and replace it with the path the Packet is going through.
- Change SF bit: It signals that the protocol changes the SF between nodes. Section 3.4 details the uses of the Debug hop and the Change SF bit.

3.3. Communication scheme

3.3.1. Point-to-point mode

The protocol follows a request/reply scheme from the *Requester Node* to a *Source Node*. The *Requester* sends commands to the *Source* and waits for the reply. If no answer is received within a certain interval or the message is corrupted, the *Requester Node* repeats the request until the reply is correctly received. Fig. 6 shows the message exchanged between the nodes that are divided into four phases: Establish Connection, Ask for metadata, Ask for data, and Final Acknowledgment.

The communication begins with the *Digital Endpoint* in the **Establish connection** state. During this phase, the *Requester/Gateway Node* sends an OK command to the *Source Node* and waits for a corresponding OK message in return. The *Requester* will wait for this response for a fixed period, known as a timeout. If a message different from the expected reply (i.e., OK) is received, it is discarded, and the *Requester* continues to listen for the remaining time. Once the OK message is received, the *Digital Endpoint* proceeds to the following three states in a loop.

In the **Ask for metadata** state, the *Digital Endpoint* requests the metadata of the content to be received from the *Source*. It waits until a packet arrives with the name and the number of file chunks. At this stage, the *Digital Endpoint* creates an empty *AlLoRa* File object that will act as a container for the incoming chunks. If this process is successful, the system moves on to the next state. During the **Ask for data** state, the *Requester* asks for chunks in sequential order to obtain the full content. Each chunk is added to the *File* object as it arrives. Once the *File* is complete, it is reassembled and ready to be accessed or saved in the *Digital Endpoint*. Finally, in the **Final acknowledge** state, an OK command is sent to maintain synchronization between nodes. The system waits until the *Source* replies with an OK command. Once this is completed, the *Digital Endpoint* returns to the “Ask metadata” state, preparing to receive the next *File*.

3.3.2. Mesh mode

The mesh mode of the *AlLoRa* protocol is designed to extend the communication range. This is achieved by utilizing intermediary nodes that forward packets until they reach their intended destination. This mode is activated when a request to a *Source* node remains unanswered after a certain number of attempts, as specified by the user.

In mesh mode, the communication process largely follows the same procedure described in the point-to-point mode (Section 3.3.1). However, when a *Source* node fails to respond to a request after a certain number of attempts, the *Digital Endpoint* associated with that *Source* node shifts into a “retransmission mode”. In this mode, the *Digital Endpoint* activates the Mesh bit in its packets, signaling the need for retransmissions.

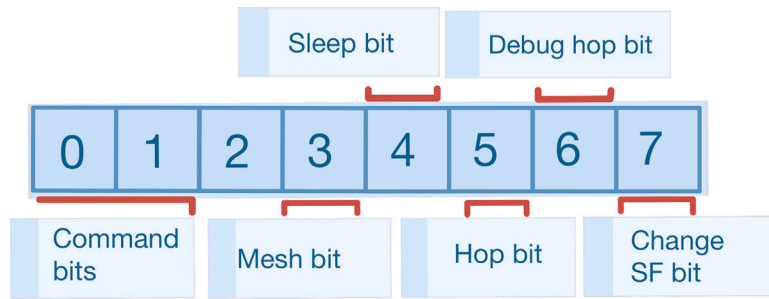


Fig. 5. The Flag Byte composition.

Table 1

Command codes and their descriptions.

Command	Code	Description
DATA	00	Activated when the payload contains a requested chunk.
OK	01	Used to establish a connection between nodes or notify of the correct reception of the final chunk of the received content. It implies that the payload is empty.
CHUNK	10	Used by the Requester/Gateway to request a chunk of the received content. The chunk number is stored in the payload, so the Source can know the requested data block.
METADATA	11	Used by the Requester/Gateway to ask for the metadata of the File to be received. If this is the case, the request's payload will be empty. The Source also uses it to answer the metadata request. In this case, the payload contains the name and size of the File to be sent.

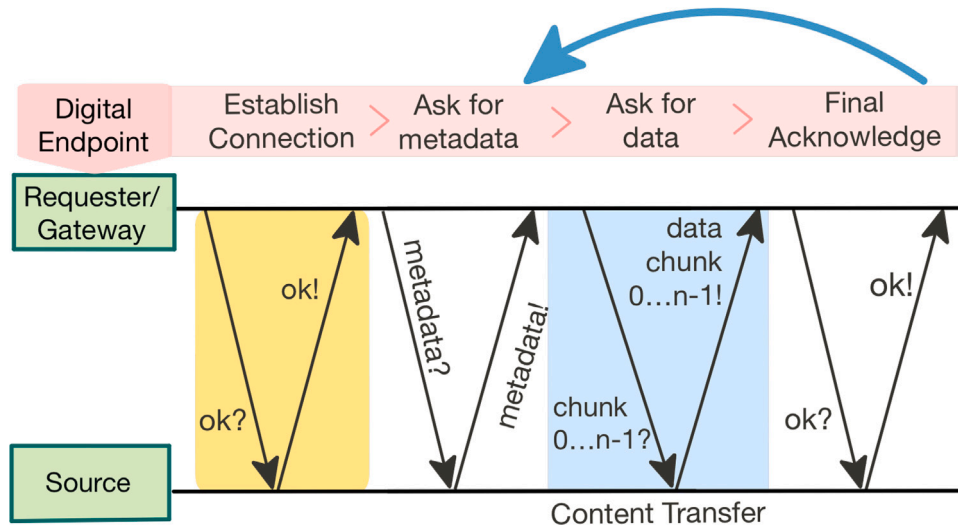


Fig. 6. Phases of the communication flow.

This shift to retransmission mode enables the protocol to maintain communication over extended ranges by dynamically adjusting to situations where direct communication with a Source node is impossible. This makes the mesh mode a powerful tool for enhancing the reach and reliability of the *ALoRa* protocol in various IoT applications.

The flowchart of the mesh mode function when sending an *ALoRa* file is presented in Fig. 7. Whenever a node receives a Packet not addressed to it, it is discarded. With the *Mesh bit* set, the node will help it reach its destination by forwarding it if it has not already done so.

To mitigate the risk of packet collisions, particularly in scenarios where multiple nodes in proximity might retransmit packets not intended for them, the *ALoRa* protocol introduces a random delay of 0.1 to 0.5 s prior to packet forwarding. This delay, controlled by the *Sleep bit*, can be deactivated in situations where the user anticipates no

collision risk. Following the initiation of a request, the Gateway node enters a waiting state for a fixed duration, during which it expects a response to arrive. If the Gateway receives a retransmitted version of its own packet from another node during this period, it disregards the retransmitted packet and continues to wait for the remaining duration for the expected response. In the event of a timeout, the Gateway continues its attempts to reach the node for the remaining focus time. Once this time has elapsed, the Gateway shifts its attention to the next node in the list of registered *Digital Endpoints*. This ensures a balanced distribution of communication attempts across all nodes.

In the *ALoRa* protocol, the activation of the *Hop bit* indicates the forwarding of a packet. This allows the destination node to discern whether the received packet has been relayed through other devices (arrived via the *Mesh mode*) or was sent directly from the source node. If a packet arrives at the destination node via intermediary nodes, the

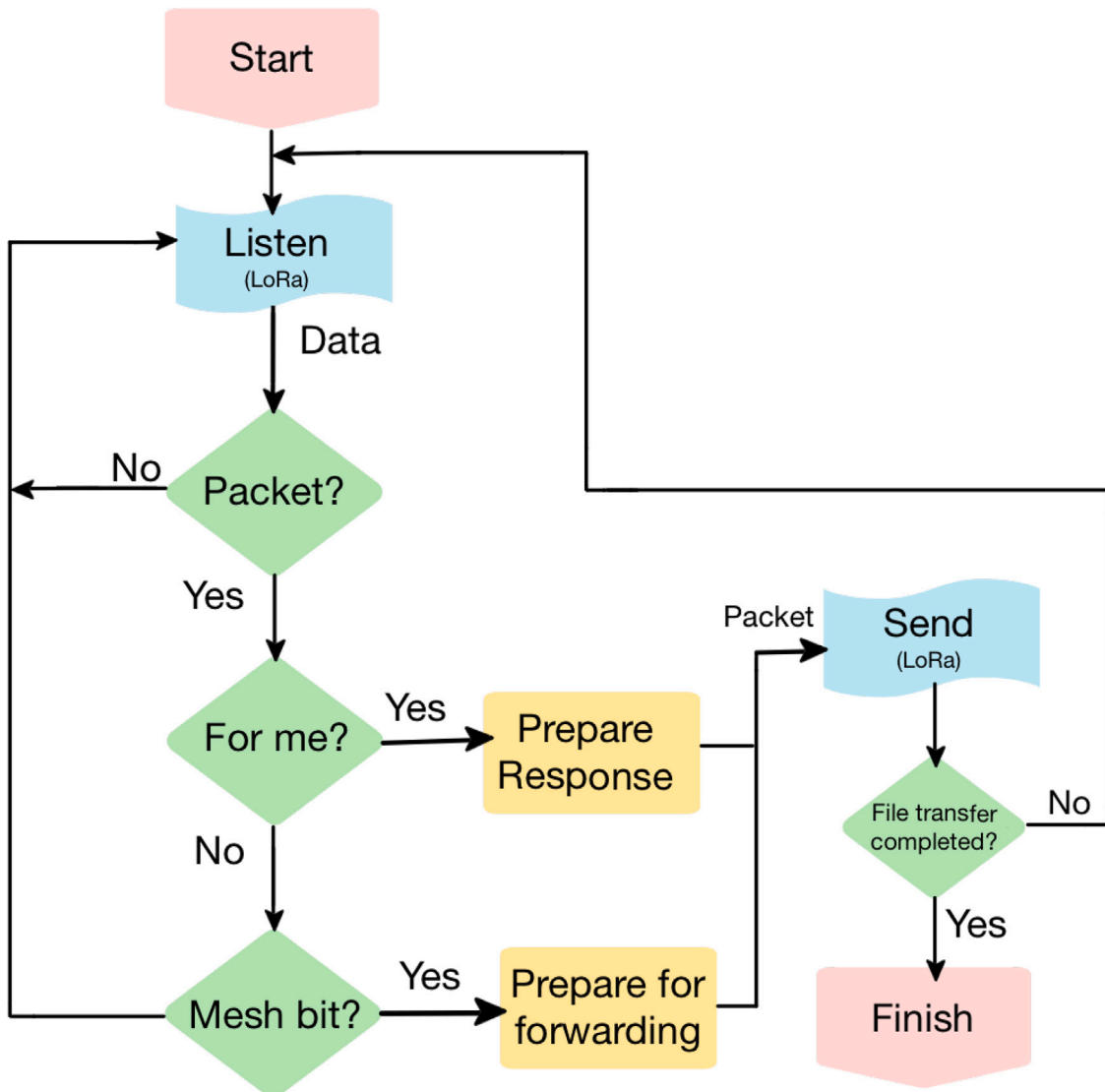


Fig. 7. Send File with mesh mode function flowchart.

Source Node generates a response packet with the *Mesh bit* activated. This is based on the assumption that the response will follow the same path back to the Gateway. If the packet arrives directly from the source node, the Source Node responds directly with the *Mesh bit* deactivated. This is because the protocol prioritizes the node that is currently being polled, ensuring timely responses to requests. If a response packet is received by the Gateway with the *Hop bit* deactivated, it signifies that the Source Node is within direct communication range, and intermediary nodes are not required to facilitate communication. In such scenarios, the Requester/Gateway deactivates the *Mesh mode* for the corresponding *Digital Endpoint*. The same action is taken if the *Mesh bit* in the response is deactivated.

To enhance the robustness and efficiency of the *ALoRa* protocol in mesh mode, a specific mechanism is employed. This mechanism assigns a unique random ID to each new packet to prevent duplication and unnecessary retransmission of messages. This ID is stored in a queue that is referenced every time a new message with the *Mesh bit* activated is received. By checking this queue, the nodes can effectively identify and disregard any duplicate packets, thereby preventing the unnecessary retransmission of messages that could potentially congest the network. In addition to this queue, the nodes maintain a second list of fixed size that records the IDs of all forwarded messages. This list is checked prior to the forwarding of any packet, ensuring that the same

packet is not forwarded multiple times. This dual-layered mechanism of assigning unique IDs to packets and maintaining comprehensive records of received and forwarded messages significantly enhances the efficiency of the *ALoRa* protocol in mesh mode, ensuring reliable communication while minimizing network congestion.

The approach utilized in *ALoRa* for collision avoidance in mesh mode is simple and with low overhead. However, special attention is required for scenarios with a high density of nodes. In the Smartlagoon project, the aim is to have the least amount of intervention to the environment, and therefore, it results in a sparse distribution of nodes. For other case studies, scalability in the face of collisions can be achieved using different radio channels for each sensor node to avoid a collision [24] or to implement a Channel Activity Detection protocol (CAD) [46]. CAD operation involves listening for a LoRa preamble, which if not detected returns an all-clear status to proceed with the transmission. Prade et al. [24] proposed a frequency multiplexing control algorithm to reduce the collision rate and data is partitioned between two available interfaces simultaneously.

3.4. Additional services

To enhance adaptability and performance in dynamic communication environments, *ALoRa* provides two additional services. The first

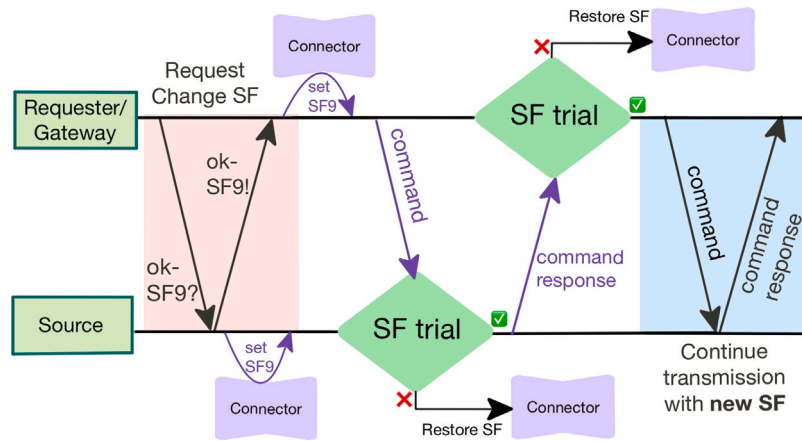


Fig. 8. Change SF flow.

is the **SF change request**, a critical feature for wirelessly adjusting the Spreading Factor (SF) value. This service allows dynamic reconfiguration of node parameters to accommodate environmental changes or application requirements without physical intervention.

The SF change is initiated by a *Requester or Gateway Node*, which sends an OK command along with the new SF value in the payload, flagged by the *Change SF bit*. Upon receiving this packet, nodes update their SF setting to the new value. If the SF update is successful, nodes continue operation with the updated configuration; otherwise, they revert to their previous SF setting after a specified number of unsuccessful communication attempts. This fallback mechanism ensures network reliability and prevents nodes from becoming isolated due to incompatible SF settings.

Fig. 8 illustrates the communication process involved in changing the LoRa SF on both sides.

If a *Source Node* receives a packet structured as previously explained, it will reply with the same structure and change the SF value for its connector to start listening to messages in the new SF configuration. Then, a trial phase starts, where the Source will attempt three times to receive messages from the Requester. If any message arrives, the changing process is completed, the new SF is settled, and a backup with this new configuration will be saved. If not, the node will revert to the original SF. The second additional service is called **Debug Mesh mode**. It is developed to test the message routes when mesh mode is enabled. Triggered by the *Requester or Gateway Node*, it activates the *Debug Hop bit* in the request packets. This feature is instrumental if a node must forward a message, as it overrides the payload and adds the “hop” information, including its name, the RSSI of the message received, and the wait time before forwarding the packet. When the packet reaches the destination node, it responds by maintaining the payload (with the route information) and adds it to a list. Upon reaching the gateway, the payload reveals the full path traversed by the message, assisting in determining the optimal distribution of nodes in an area for deploying an *AlLoRa* network.

The strategic adaptation of SF in response to the varying demands of IoT applications is crucial for network efficiency. A lower SF may be preferred to optimize data throughput in urban environments with dense node deployments. In contrast, a higher SF may be necessary in rural or expansive areas to ensure long-range connectivity. This flexibility is vital for the robust performance of the network under diverse conditions. However, it is important to calibrate these adjustments precisely. Overly frequent or poorly optimized changes in SF can lead to increased signaling overhead and energy consumption. Such inefficiencies not only strain the network resources but also can shorten the lifespan of battery-powered nodes and reduce the overall operational efficiency of the network.

4. Results and discussion

In this section, we delve into the empirical evaluation of *AlLoRa*, assessing its performance through a series of real-world tests. These tests were designed to measure the protocol’s transfer efficiency, energy usage, effective range, and mesh networking capabilities, with a focus on the adaptability of *AlLoRa* to diverse environmental conditions and its scalability for energy-efficient IoT implementations.

The prototype configuration used for these tests is depicted in Fig. 9. A Raspberry Pi 4 served as the Gateway node, with a Pycom LoPy4.¹ A device providing LoRa access via a Wi-Fi channel, functioning as an *AlLoRa* adapter. Additional Pycom LoPy4 devices were configured as *Source Nodes*.

The selection of Pycom LoPy4 for both the Gateway node and Source Nodes reflects a deliberate choice to balance cost, performance, and ease of integration. The integration of a Raspberry Pi 4 as the Gateway node is a strategic choice, capitalizing on its affordability and capacity for data processing. This enables a cost-effective yet functional gateway solution that aligns with the lightweight and efficiency-driven design of *AlLoRa*.

Tests were conducted using the European 868MHz band, with a default Spreading Factor (SF) of 7, increased to SF11 for bandwidth and energy consumption tests. The protocol’s support was extended to ESP32-based microcontrollers, specifically the Heltec Wi-Fi LoRa 32 and the LILY GO LoRa 32.

The use of a single-channel LoRa gateway, as in this implementation of the *AlLoRa* protocol, offers simplicity and cost-effectiveness, suitable for scenarios with fewer nodes and sparse network distribution. However, this configuration limits the network to a single channel, potentially restricting scalability.

Multi-channel LoRa gateways, while more complex and costly, can handle more traffic and support a larger number of nodes, suitable for dense networks. As the network grows, transitioning to a multi-channel gateway might be necessary to handle increased traffic and prevent data collisions. Therefore, the choice between single-channel and multi-channel LoRa involves a trade-off between cost, complexity, and network capacity, necessitating careful consideration of the specific network requirements and anticipated growth.

While the current version of *AlLoRa* does not consider multi-channel gateways, the protocol is designed in a way that is compatible with such a configuration. This means that the implementation of multi-channel gateway support could be easily integrated into future versions of the protocol, further enhancing its scalability and adaptability to diverse network conditions.

¹ <https://pycom.io/>

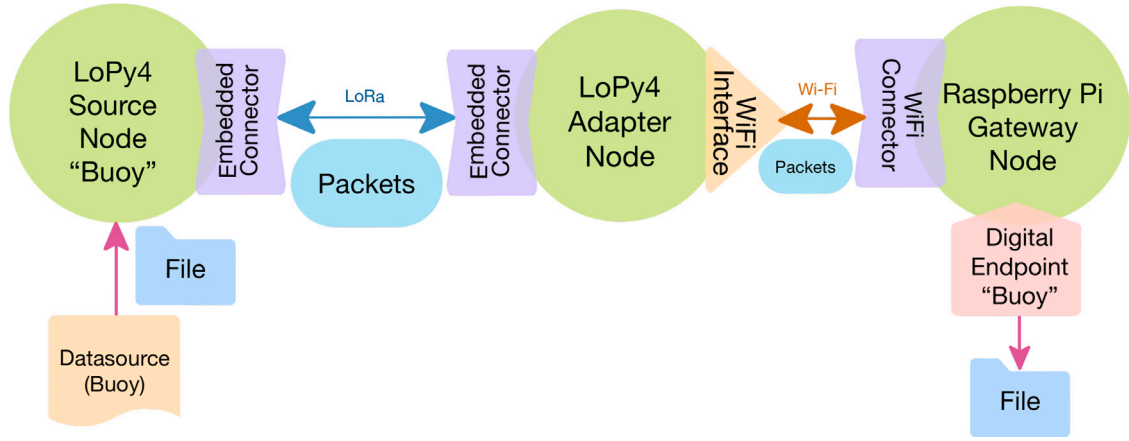


Fig. 9. The modules distribution and hardware used for the prototype used.

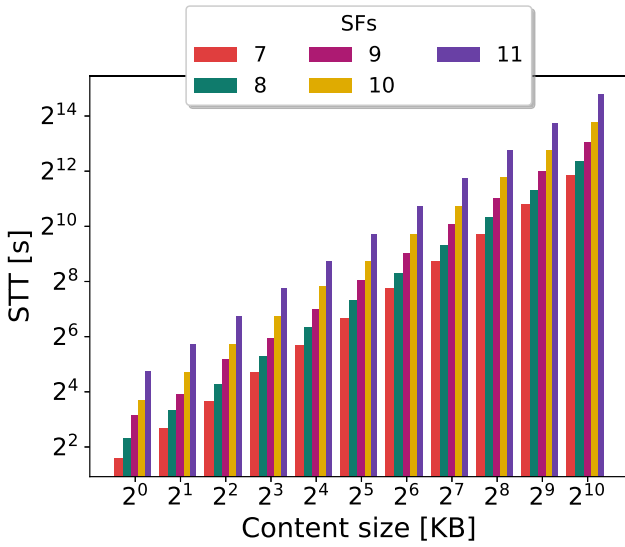


Fig. 10. STTs per content size, varying the SF.

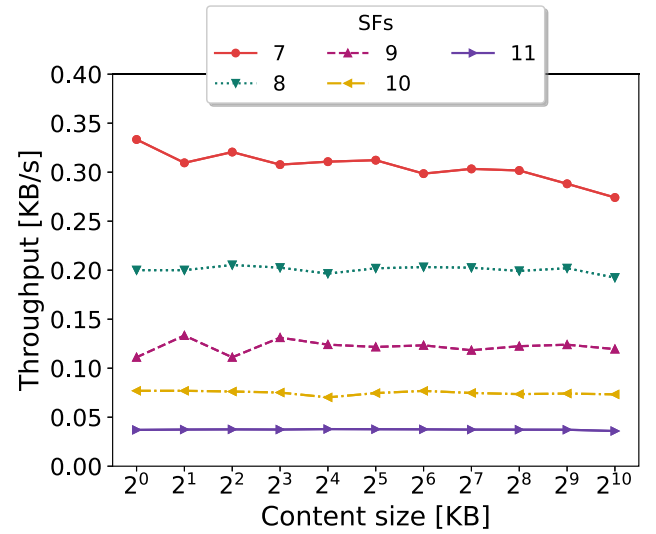


Fig. 11. Average throughput per content size varying SF.

4.1. Transfer time tests

The “Successful Transfer Time” (STT) is a crucial metric for assessing the efficiency of *ALoRa*. It encapsulates the time required for a *Source Node* to successfully send a message, including both the data transmission and the acknowledgment reception. This metric is fundamental in estimating the throughput of the communication system, providing insights into the protocol’s performance under various conditions.

The distance between the deployed nodes was 1 meter. The *Source Node* was set to send increasingly larger messages from 1 to 1024 kilobytes (1 MB), with the STT recorded for each transmission. Additionally, the SF value was varied from 7 to 11 to examine its impact on the time and throughput, using a chunk size of 235 bytes, the maximum allowable with *ALoRa* when mesh mode is disabled (see Fig. 3).

Fig. 10 shows the value of the STTs by varying the file size and varying the SF. Sending 1 KB took about 4 s with SF7, and 27 s with SF11, while sending 1 MB took about 1 h with SF7 and about 8 h with SF11.

Fig. 11 shows the estimated throughput for each content size tested while also varying the SF. The throughput varies therefore from around 256 Bps (approximately 2 kbps) with SF7 to 38 Bps (approximately 300 bps) with SF11 and is stable and linear with the increase of the content size and the SF. As an example, the Mar Menor Buoy’s data

logger generates contents of 33 kB each, requiring an average transfer time of 100 s.

The data presented in Figs. 10 and 11 illustrate the influence of SF on transmission times and throughput. With higher SF values, STT increases due to extended airtime required for the same data volume, a factor that could impact the protocol’s operational efficiency, especially over larger distances. This test validates the importance of SF selection in relation to the network environment, node density, and distance coverage, providing a basis for configuring *ALoRa* in diverse application scenarios.

Furthermore, the relationship between SF and throughput is crucial for practical deployments. The results indicate a decrease in throughput by approximately sevenfold when switching from SF7 to SF11, underscoring the need for a balanced approach in SF tuning to optimize both range and data transmission rates. These findings will guide users in adapting *ALoRa* to their specific needs, ensuring reliable data communication while maximizing network longevity.

4.2. Energy consumption test

Sustainability in IoT networks is closely tied to energy efficiency, particularly in scenarios where power resources are scarce or replenishment is challenging. Therefore, we closely scrutinized the power requirements of *ALoRa*, aiming to optimize energy consumption without compromising communication effectiveness.

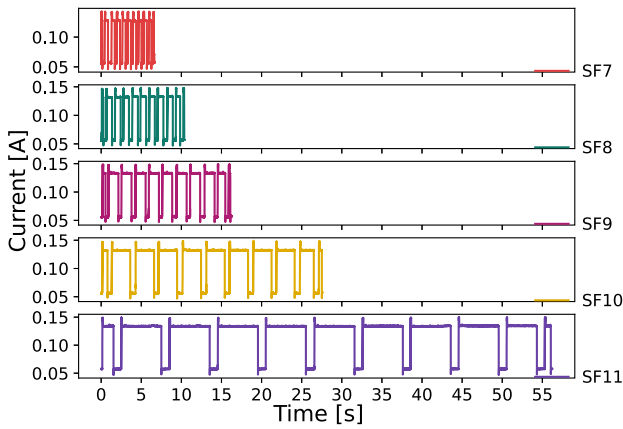


Fig. 12. Current vs. time, sending 2 kilobytes of content at different SFs.

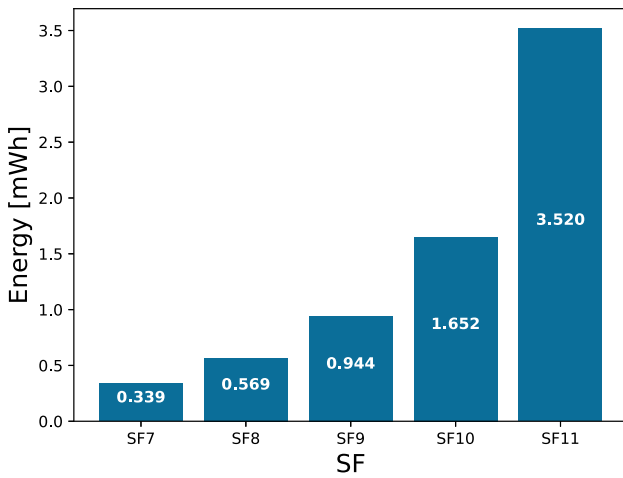


Fig. 13. Energy to send 1 KB per SF.

This section shows the evaluation of the power consumption for sending files of different sizes. The Source node is connected to a current and voltage measurement unit.² We repeated the same experiments of Section 4.1.

Fig. 12 shows the current usage behaviors obtained while a LoPy4 was sending 2 KB of content with each tested SF. It is interesting to analyze this graph, as each pulse represents a phase of communication, as shown in Section 3.3. The first pulse is the response for metadata, the next 9 pulses are the data chunks of the content being transferred. The ninth pulse seems to have a smaller width than the rest; this is because 8.79 chunks are needed to send 2 KB of data, so this last pulse contains 0.79 of the remaining data. The last pulse is the final acknowledgment after the Requester obtains all the chunks of the file being transferred. This figure helps visualize once more the impact of the SF used on the duration of the data transmission and therefore on the power demand of the system.

Fig. 13 shows the energy needed to transmit 1 KB for increasing values of SF. The average consumption using SF7 was 339 $\mu\text{W} \cdot \text{h}$ per kilobyte sent, increasing to 3.52 $\text{mW} \cdot \text{h}$ per kilobyte when transmitting with SF11. This result was expected because of the longer time-on-air needed with a higher SF.

We also tested the average power consumption per KB with SF7 of *ALoRa* using the ESP32 devices Heltec Wi-Fi LoRa 32 and the LILY GO LoRa 32. For the Heltec device, we obtained a value of 428 $\mu\text{W} \cdot \text{h}$ per

KB, while with the LILY GO LoRa 32 the average power consumption was 395 $\mu\text{W} \cdot \text{h}$ per KB transmitted. These variations are minimum and clearly depend on the specific board being used.

The data on energy consumption provide actionable insights for field deployments, where the longevity of sensor nodes is crucial. By adjusting the SF, *ALoRa* demonstrates the capability to balance energy usage against the need for range, which is vital for remote monitoring activities.

4.3. Range test

The effective communication range is a vital metric for environmental monitoring applications, particularly over extensive areas such as the Mar Menor lagoon. The range tests conducted in different terrains validate *ALoRa*'s robustness and its capability to maintain reliable connections over considerable distances.

The range test in Mar Menor was done using a *Source Node* sending a small message simply containing the Received Signal Strength Indicator (RSSI) of each request received. The gateway was installed on the roof of the IEO (Spanish Institute of Oceanography, Murcia, Spain)³ at a height of approximately 25 m. The *Source Node* was moving in a boat toward the predetermined buoy coordinates. We registered the GPS position of the *Source Node* to compare the RSSI value with the Gateway distance. We managed to communicate from the *Source Node* to the *Gateway* at a maximum distance of 11.69 km, achieving the intended location of the buoy (see Fig. 14).

We repeated the same range test at Erken Lake, Sweden, and achieved approximately 8.4 km of communication between the nodes, as shown in Fig. 15. In this case, the gateway antenna was approximately 4 meters in height, which, combined with the number of trees and changes in terrain, may explain the decrease in range compared to the test in Mar Menor.

These tests, conducted in varied geographic settings, from the coastal Mar Menor to the wooded areas surrounding Erken Lake, demonstrate the protocol's versatility and confirm its suitability for a broad spectrum of deployment scenarios.

4.4. Mesh mode tests

Mesh networking in *ALoRa* offers enhanced flexibility and operational reach, particularly valuable in complex or challenging network environments. The following tests, conducted within the Mar Menor context, were designed to evaluate the functionality and reliability of this mesh networking feature.

The initial feasibility test involved disrupting the direct link from the gateway to the buoy. An additional *ALoRa* node was strategically placed on a mountain, 5.5 km away from the buoy and at an elevation of 65 m, to act as a forwarder. This test setup was critical in evaluating the mesh mode's capacity to maintain connectivity and effectively manage data forwarding in scenarios where direct communication paths are obstructed.

In Fig. 16, the Gateway is represented with the letter G, the Buoy Node is represented with the letter B, and the forwarder Node with A. Between G and A there was a distance of 14.78 km, but with A being placed at a higher altitude than B, the Gateway messages were constantly arriving.

We started by replicating the Gateway at the location A. Establishing communication directly to the Buoy and receiving its messages as expected (point-to-point communication).

Then, we configured Node A as a *Source Node*. When activated, it began receiving and forwarding requests from G intended for the Buoy, and also receiving the Buoy's replies and forwarding them to G, achieving a mesh with a total route length of 20.33 kilometers. This experiment allowed us to validate the mesh forwarding approach.

² Otili Arc: <https://www.goitech.com>

³ <http://www.ieo.es/es/web/murcia/>

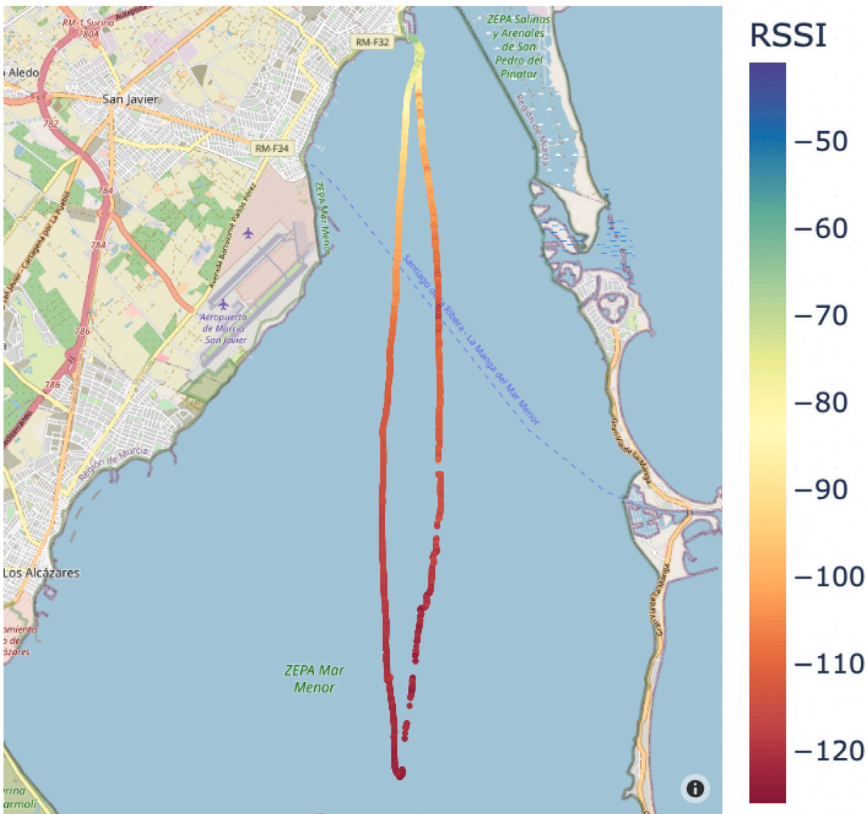


Fig. 14. RSSI Map distribution in Mar Menor, Murcia, Spain.

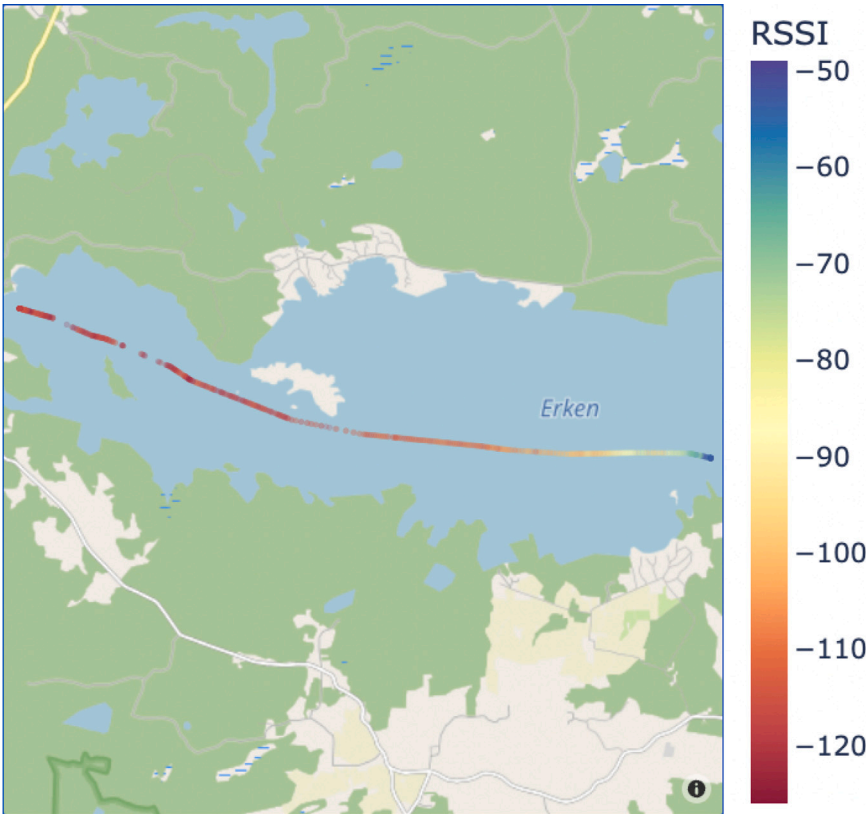


Fig. 15. RSSI Map distribution in Erken Lake, Sweden.

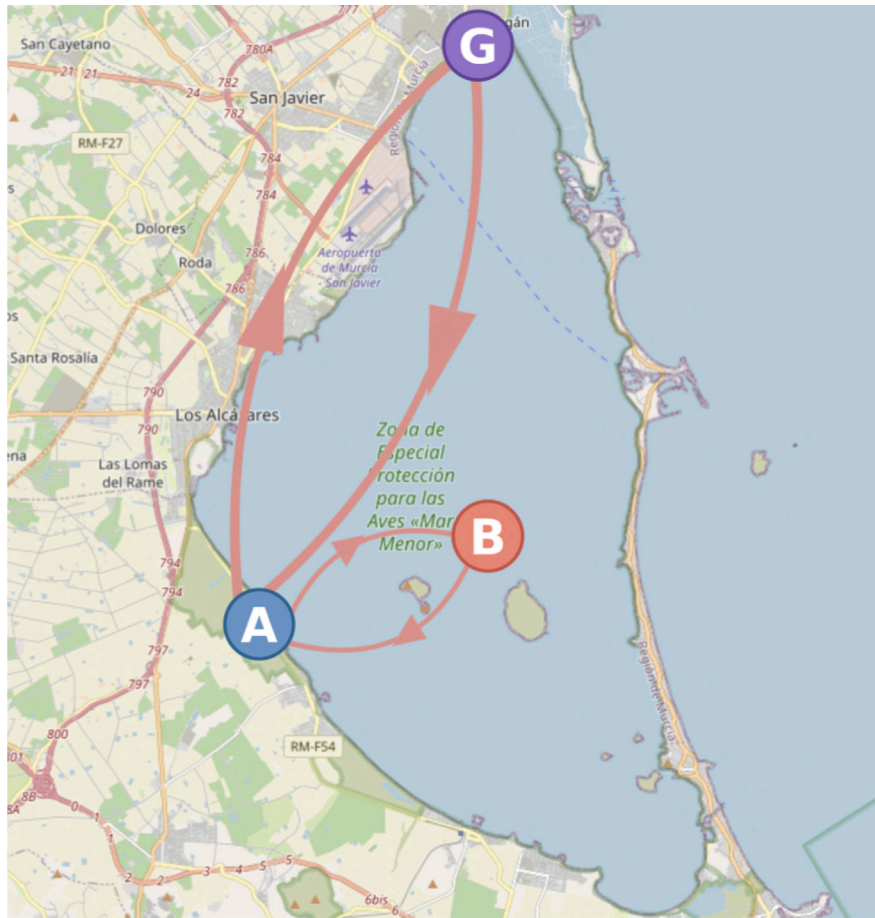


Fig. 16. The Mar Menor Mesh feasibility test.

To conduct a more thorough study, we set up three nodes as Source (labeled A, B, and C, respectively) and set up a Gateway (G) in “debug hop mode” with “sleep mesh” off. Source nodes were distributed in different configurations and distances in the *Universitat Politècnica de València*’s Campus to test the path followed by the Packets. We tested it under challenging conditions for LoRa, without *line of sight*, many objects and buildings, and with the nodes approximately 1 m from the ground, forcing low signals to evaluate the system’s behavior with forwarding enabled. The Gateway was configured to ask for data only to node A, to evaluate the differences in packet reception rate for the furthest Node in each arrangement. Each case was tested for 15 min.

To evaluate and visualize the Mesh mode, we developed an arch diagram that represents node interactions, where the curves’ color and style denote the message’s intended recipient, and their thickness reflects the number of messages that have traveled along that path (with thicker lines portraying more messages), and their opacity indicates the message’s RSSI, with more visible indicating a stronger signal. The message’s direction is represented with arrowheads. Hence, forwarded messages can be recognized when a node receives a curve not intended for it, and curves with the same destination can be seen leaving it.

We tested the nodes distributed in 3 configuration types. First, to validate the set-up, we located Node A about 200 meters from the Gateway (see Fig. 17). As expected, the Gateway received correct messages from the node, as it works precisely like the point-to-point configuration. The average RSSI registered in this case was -95 dBm. This location was selected to be the point where even a small distance increase produced a drastic decrease in the RSSI and packet losses became considerably high, due to the presence of a building. In this attempt, the average response packet reception was of about 3 per second.

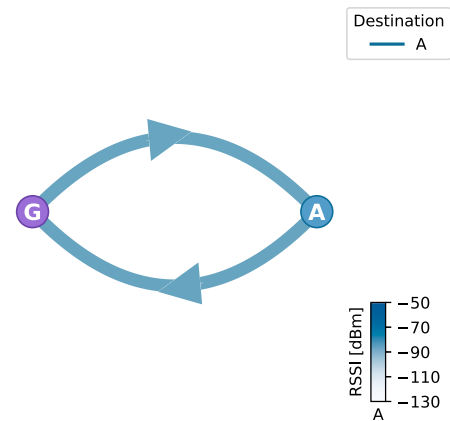


Fig. 17. Nodes’ distribution for *mesh mode* test: first case with a single hop.

In the second configuration (see Fig. 18), we replaced Node A with B and increased the distance to force node A to lose direct contact with G. At that point, the Gateway activated the *mesh mode* for Node A and B started acting as a forwarder node. Finally, we placed Node A 200 m from B. The average RSSI registered between A and B was -80 dBm, and the average response packet reception was about 2 per second. As can be seen from Fig. 18 the majority of messages for Node A were forwarded through Node B, but in some cases, point-to-point communication between A and G was still achieved, as the tested distances were not high enough for LoRa. This is a realistic event, since propagation can vary during time, and it is handled properly by *ALLoRa*.

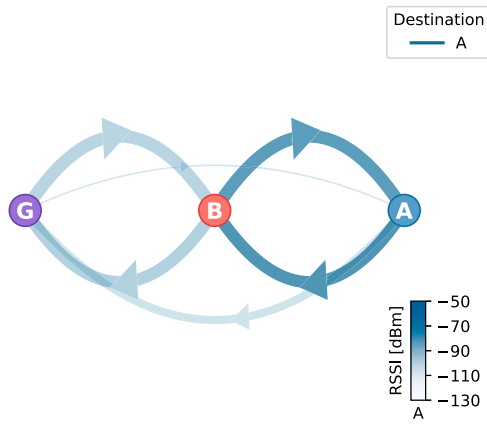


Fig. 18. Node's distribution for Mesh mode test: the second case.

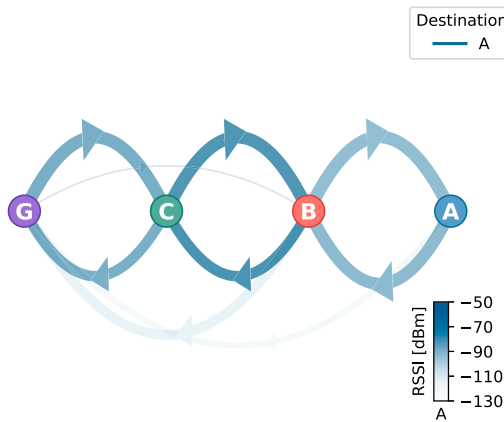


Fig. 19. Nodes' distribution for Mesh mode test: the third case.

Finally, in the third configuration, we placed Node C in B's place at the first 200 m, replaced Node A for B at 400 m, and then went another 200 m away with Node A. The average RSSI registered between B and A in this case was of -93 dBm. As shown in Fig. 19, with this distribution of nodes, the messages directed to A were "hopping" 2 times, first being forwarded by node C and then by Node B. A's responses followed the reverse path, eventually arriving at the Gateway as responses for its original requests. In this case, the response packets arrived at about 1 per second, a third of the base case. Same as with the last case, it can be observed that some packets went directly between the Gateway and B, and also some packets arrived directly from Node A to G.

With this nodes' distribution, we triplicated the original distance of the point-to-point set-up with improved speeds than those obtained with the original mesh test in Mar Menor. Along these experiments *ALoRa* maintained a stable functioning behavior, properly handling the connectivity tasks and maintaining the data transfer performance.

These tests showcase *ALoRa*'s ability to adaptively use mesh networking for consistent network performance, even when direct line-of-sight communication is obstructed or impractical. This feature proves particularly useful in maintaining network integrity in diverse and challenging environmental conditions.

4.5. Comparative analysis with other IoT communication technologies

This section proposes a comparative analysis of *ALoRa* against possibly equivalent technologies such as the LoRa modulation technique, the LoRaWAN network protocol, and the Bluetooth and Wi-Fi wireless standards, all of which are accessible through the tested devices.

Additionally, we consider the broader context by including cellular networks and satellite technologies, which, though not directly accessible through our test devices, play a significant role in IoT communications. Drawing from our *ALoRa* test results, we aim to highlight its potential advantages and suitability for several IoT applications.

It is important to note that technologies like LoRa, LoRaWAN and *ALoRa*, while offering more extended range and lower power requirements, generally provide slower data transmission speeds compared to Bluetooth, Wi-Fi, and cellular networks. This trade-off is crucial in contexts where the range and energy efficiency are prioritized over high-speed data transmission.

Starting with range capabilities, *ALoRa* demonstrates a distinct advantage over technologies such as Bluetooth and Wi-Fi. While Bluetooth and Wi-Fi generally offer shorter-range communications (up to a few hundred meters), *ALoRa* achieved effective long-range connectivity of up to 11.69 kilometers using point-to-point communication (as observed in 4.3), making it particularly advantageous for remote environmental monitoring.

While standard LoRa provides the basics for point-to-point communication, it lacks the comprehensive features offered by *ALoRa*. The protocol enhances LoRa's basic functionality with a modular design, facilitating adaptable deployments in various scenarios. This approach is further enriched by advanced features such as mesh networking and efficient data handling, contributing to a robust and user-friendly experience. Moreover, *ALoRa* excels in node and file management, offering a systematic approach to handling complex network configurations and data flows. This makes *ALoRa* significantly more suitable for diverse IoT applications than using the LoRa physical layer in standard configurations.

LoRaWAN, designed for end-to-end wide-area network coverage, is not suitable for direct point-to-point connections with nearby IoT nodes. Moreover, inheriting limitations from its underlying LoRa technology, it faces certain constraints. For example, while *ALoRa* demonstrates the ability to transmit 1 KB of data in about 4 s using SF7, as depicted in Fig. 10, achieving this in LoRaWAN would require the "manual" transmission of multiple payload fragments from the device. Upon receipt, these fragments must be reassembled, rendering the use of LoRaWAN for such tasks significantly more challenging.

Shifting focus to cellular networks like 4G and 5G or even to satellite technologies like the emerging low Earth orbit (LEO), it is important to contextualize their role in IoT communications. While these technologies offer high-speed data transfer and extensive coverage, their reliance on established infrastructure and ongoing costs associated with data plans and network services limit their applicability in remote or undeveloped areas. This makes them less viable for environmental intelligence projects, especially in cost-sensitive or infrastructure-limited scenarios.

From all the above, *ALoRa* emerges as a versatile and efficient communication protocol that excels in scenarios where range, flexibility, and cost-effectiveness are paramount. Its performance in our tests highlights its suitability for environmental monitoring and IoT applications, where it can reliably operate under varying conditions without the constraints of more complex and costly technologies.

5. Conclusions

In this work, we introduce *ALoRa* a high range, low power, and modular communication protocol, based on LoRa, designed explicitly for environmental intelligence-oriented deployments. Our solution is energetically sustainable and flexible enough to adapt to a variable number of sensing devices and modular to allow additional hardware implementations to be used in the deployment.

An exhaustive evaluation was conducted with a real test bed in a 135 km² coastal lagoon, where we developed an oceanographic monitoring system. We showed that it offers the required performance according to transfer time, energy consumption, range, and performance in mesh mode.

As future work, we expect to analyze the system's robustness and scalability using a larger scale deployment in Mar Menor. The objective is to communicate a larger number of sensors that capture a more diverse set of environmental metrics. Future developments may also apply *AlLoRa* to other large-scale real applications such as smart cities, tourism, and agriculture.

Further research is required to provide an efficient solution in the context of mobile sensors, hence, we plan to include dynamic SF capabilities to adapt to different conditions. Moreover, security and privacy will also be considered in future implementations of *AlLoRa*.

CRedit authorship contribution statement

Benjamín Arratia: Conceptualization, Data curation, Investigation, Methodology, Software, Validation, Visualization, Writing – original draft. **Erika Rosas:** Data curation, Investigation, Validation, Writing – original draft. **Carlos T. Calafate:** Conceptualization, Formal analysis, Validation, Writing – review & editing, Supervision. **Juan-Carlos Cano:** Formal analysis, Funding acquisition, Project administration, Supervision, Validation, Writing – review & editing. **José M. Cecilia:** Conceptualization, Funding acquisition, Methodology, Project administration, Resources, Supervision, Writing – original draft, Writing – review & editing. **Pietro Manzoni:** Conceptualization, Formal analysis, Funding acquisition, Investigation, Methodology, Project administration, Resources, Supervision, Validation, Writing – original draft, Writing – review & editing.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Data availability

Data will be made available on request.

Acknowledgments

This work has been supported by the European Union's Horizon 2020 research and innovation program under grant agreement No 101017861 and also by the project Ramon y Cajal Grant RYC2018-025580-I, funded by MCIN/AEI/ 10.13039/501100011033 and ESF Investing in your future, GVA GRISOLIAP/2021/103 and Grant TED2021-130890B funded by MCIN/AEI and by European Union NextGenerationEU/PRTR.

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